

On the minimum dataset size and quality for a maize seedling object detector

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Keywords: keyword 1; keyword 2; keyword 3 (List three to ten pertinent keywords specific to the article; yet reasonably common within the subject discipline.)

1. Introduction

Plant counting is a critical operation in precision agriculture, plant breeding, and agronomical evaluation. Accurate plant counts can provide valuable information for both farmers and researchers. This task was often performed manually by human operators. Now a days this process can be automated by the use of computer vision algorithms. To validate a method for counting, it is critical to set a benchmark for accuracy. A benchmark can be defined by the accuracy of manual counting, international standards, or by comparison with other methods. Accuracy of plant manual counting depends on human performance, so difficult to define, but it is often taken as golden sample. According to the European Plant Protection Organization, the benchmark for acceptance is a coefficient of determination (r squared) of 0.85 when compared to manual counting [1]. This corresponds to a Root Mean Square Error (RMSE) of approximately 0.39. Also scientific literature mention a r -squared value of 0.85 with ground truth (manual counting) as a benchmark for acceptance [2]. Literature shows that benchmark for acceptance can be achieved with computer vision object detection [3–5] or regression models [6,7]. A superiority of object detection over regression models in terms of accuracy was found [2]. Object detection is also more versatile than regression models, because, object detection model inference on georeferenced orthomosaics can be used to verify plant presence through its coordinates throughout the vegetative phases. This is not possible with regression models, because they can only predict the number of plants in an image.

The ability to identify and locate the objects within an image are the main features of object detection models. The validation of this ability rely on metrics as Intersection

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over Union (IoU), Average Precision (AP), and Average Recall (AR) [8] rather than the coefficient of determination. Identification supports are usually bounding boxes, which are rectangles that enclose the object of interest, but they can also be points or other kind of polygons. Counting of plants is then commonly performed by counting the number of bounding boxes that enclose the objects of interest (the plant) in a image area. Georeferenced orthomosaics are images created through aerial photogrammetry, a process that involves capturing overlapping aerial georeferenced images taken by nadiral view (and/or picturing georeferenced ground control points), and performing bundle adjustment to form a single, seamless image [9]. Georeferenced orthomosaics are scaled and oriented to a geographical coordinate system. It implies that the pixel coordinates and size correspond to geographical coordinates and are projectables in a metric system. Using georeferenced orthomosaics in seedling counting makes object detection easier, because the metric scale can be used to locate the objects in the image and prospective variance is reduced to zero.

The development of object detection algorithms has evolved from not machine learning methods, here named handcrafted methods (HC), to modern deep learning-based (DL) techniques. Pratically all the state-of-the-art object detection methods are based on DL models now a days. Nethertheless, HC methods are still used in some cases [4]. Modern DL object detection uses CNN-based frameworks (e.g., Faster R-CNN [10], YOLO [11]) and Transformer-based approaches (e.g., DETR [12]). The main difference between CNN-based and Transformer-based models is the way they process the image. CNN-based models process the image in a grid-like fashion, while Transformer-based models process the image as a sequence of patches. On common benchmarks as COCO [8], PASCAL VOC [13], and ImageNet [14], Transformer-based models have shown to be more accurate than CNN-based models [15]. Nethertheless CNN-based models are still widely used in object detection, because they are more efficient in processing small images and have a lower computational cost [16]. One of the critical challenges in training effective DL object detection models is the availability of a sufficiently large and high-quality dataset. Recently, zero-shot and few-shot object detection have emerged as promising paradigms to alleviate the need for large annotated datasets. While traditional object detection models (many-shots object detectors) require extensive labeled data for training, few-shot and zero-shot object detection aim to detect novel objects with little to no labeled examples respectively. The term "shot" refers to the number of annotations (plants for this study) used to train the model over all the images. Zero and few-shots approaches often leverage feature transfer or meta-learning components to generalize across classes under extreme data scarcity [17]. This approach reduces the annotation burden and is especially beneficial in domains where collecting exhaustive training data is impractical. Zero-shot object detection detects new categories without any training samples by leveraging semantic relationships or contextual information learned from known classes [18]. Few-shot approaches optimize models to learn quickly from a handful of labeled examples [19]. Meta learning is a common approach in few-shot object detection [20–23], while Cross-Domain Few-Shot Object Detection (CD-FSOD) has recently surpassed this approach by leveraging domain adaptation techniques [24]. Zero-shot actual state-of-the-art object detector models are YOLO-World [25], OWLv2 [26], and Grounding DINO [27]. DE-ViT [28] and CD-ViT [29] are the latest models that have shown promising results in few-shot object detection.

Many studies have been conducted on object detection for plants in open field [3,5,6,30–39], but few have focused on dataset minimum requirements for training a robust plant object detector [4,40]. Some study focused on few-shots approach for plant counting CiteHere(One to All: Toward a Unified Model for Counting Cereal Crop Heads Based on Few-Shot Learning) CiteHere(AdaTreeFormer: Few shot domain adaptation for tree counting from a single high-resolution image) but only two specifically accounting on few-

shots method for maize seedling counting [41] CiteHere(Advancing Image Recognition: Towards Lightweight Few-shot Learning Model for Maize Seedling Detection). These last set the few-shot benchmark for maize seedling counting at 600 examples. No zero-shot benchmark for maize seedling counting has been set yet, and no research has been done yet.

Even if no research has been made in order to minimize or set a benchmark for the dataset size and quality required to train a maize seedling object detector, it is well known that the performance of a DL model is directly related to the amount of data used for training [42] and its quality [43]. Dataset size and quality predictability in DL has been proven to be addressable with empirical approaches [44,45]. As already mentioned, model architecture is critical in dataset requirements as different models may require different dataset sizes and qualities to achieve the same performance [46,47]. Backbone is pivotal on downstream tasks as object detection [48]. The importance of using a domain specific backbone has been proven also for plant/leaves segmentation on orthomosaics [49], but backbone training is still prohibitive for many applications. No backbone weights specialized on agricultural orthomosaics has been published yet, as for the most of the specialized domains. So, because of limited resources, only the use of a general backbone is possible for this study, even if a decreasing in dataset size requirements is expected with a domain specific backbone [49].

Nevertheless, the most important factor affecting the dataset size for a plant counting object detector is the dataset source [4,40]. It has been observed that using training samples from the same dataset as the inference (in-domain dataset) increases accuracy and reduce the need for a large dataset in respect to collecting training samples from other datasets (out-of-distribution dataset) [4,40].

Plant counting is affected as many DL application fields from data scarcity: public datasets are rare and often not suitable for the task because of the large environmental variability CiteHere , their images are not orthorectified or scale is unknown CiteHere . Even so we can find some datasets that can be used for training a plant counting object detector CiteHere . We can reduce the variability that the model should learn by constrain the dataset to a specific crop at a specific growth stage. For this study we choose maize seedlings (*Zea mays* L.) at the V3-V5 (BBCH 13-15) growth stage. It is a common crop that in this particular growth stage is easy to count because of low overlapping and fixed intra-row and inter-row spacing.

This study aims to determine the minimum dataset size and quality required to train a benchmark reaching object detection model for identifying maize seedlings in georeferenced orthomosaics, where the dataset size and quality are respectively defined by the amount of annotated images in the training set, and the accuracy of the annotations. Also the effect of training dataset source will be evaluated in this study. Models architecture and size will be taken into account, as long as the object detection downstream task is concerned. After setting the dataset size and quality minimum requirements for many-shots object detectors, we will evaluate if new zero-shot and few-shot object detection models can achieve the same benchmark for acceptance with less data and annotations. We will also discuss the need for HC method in a DL object detection pipeline.

2. Materials and Methods

In order to investigate the minimum size and quality of the dataset required to train a robust object detection model for maize seedlings, we conducted a series of experiments where we recursively fit a series of models with increasing dataset size and quality. In this section we will first define the datasets used in the experiments, divided by in-domain and out-of-distribution training datasets. Then we will describe the architectures chosen to

represent the many-shots, few-shots, and zero-shots approaches. Finally we will describe a HC object detection algorithm to evaluate it as a strategy to get the training datasets.

2.1. Datasets

The datasets used in this study to train the object detection models for maize seedlings counting are nadiral or supposedly nadiral images of maize seedlings at the V3-V5 growth stage. The V3-V5 growth stage is defined by the BBCH scale as the stage where the third to fifth leaf is unfolded and the plant is 15-30 cm tall [?]. This growth stage is easy to determine, even from above because of the low overlapping of the leaves.

This study uses two dataset sources as training sets: the out-of-distribution dataset (OOD) and the in-domain dataset (ID). The ID datasets are from the same source as the testing dataset, while the OOD datasets are not. The OOD datasets are composed of images from scientific literature [4,6] and from internet repositories [50,51]. The ID datasets were collected during this study. The OOD scientific datasets and the ID datasets are composed by patches of georeferenced orthomosaics of known scale, while the OOD internet datasets are simple RGB images of unknown scale. The all OOD datasets came with annotations, while the ID datasets were manually annotated.

Table 1. Summary of Datasets Used in the Study

Dataset	Description
OOD Scientific	Composed of tiles of georeferenced orthomosaics of maize seedlings from scientific literature [4,6].
OOD Internet	Composed of RGB images of maize seedlings from internet repositories [50,51].
ID	Collected during this study, composed of tiles of georeferenced orthomosaics of maize seedlings of known scale.

To make the two kind of dataset comparables we chose to rescale the images to a scale of 0.005 m/pixel where the scale was known (scientific OOD and ID datasets), obtaining orthomosaics of different sizes. All the datasets were then cropped to 224*224 pixels tiles. Also the annotations were rescaled where needed and cropped.

The testing dataset is composed by a subset of the ID dataset tiles, not included in the training dataset.

2.2. many-shots object detectors

To represent the category and compare performance between pure-CNN and Transformer-mixed architectures that are effectively used as object detectors in real use-cases for their accuracy and efficiency, we chose to use three versions of YOLO and the real-time version of DETR in the Ultralytics implementation. The YOLO versions are YOLOv5, YOLOv8, and YOLO11, while the DETR version is RT-DETR.

YOLOv5 CiteHere is a pure CNN-based architecture that represents a significant optimization over previous YOLO versions with improved speed-accuracy tradeoffs and enhanced training stability. It employs a CSPDarknet53 backbone that uses cross-stage partial connections to reduce computational costs while maintaining feature representation power. YOLOv8 CiteHere introduced architectural modifications including a decoupled detection head, a C2f module for feature processing, and several advanced augmentation techniques, resulting in better performance on small objects like seedlings. Its backbone is an improved CSPDarknet with enhanced SPPF (Spatial Pyramid Pooling - Fast) modules for multi-scale feature extraction. YOLO11 CiteHere represents the latest evolution with significant improvements in accuracy through adaptable block and module designs, adaptive head assignments, and enhanced loss functions for optimizing detection at various scales.

YOLO11 uses an advanced adaptable backbone with improved feature pyramid networks for efficient multi-scale detection. RT-DETR CiteHere is a real-time transformer-based detector that combines CNN backbone features (ResNet-based) with a streamlined transformer decoder, striking a balance between the global reasoning capabilities of transformers and the efficiency of CNN architectures, making it particularly effective for detecting objects with varying sizes and densities in complex agricultural scenes.

We used here the Ultralytics implementation of these models because the implementation is open-source CiteHere and it is widely used in the scientific community CiteHere.

MakeTableHere

2.3. *few-shots object detectors*

Differently from the many-shots approach, the few-shots one was not deeply studied to tackle the problem of plant counting. In fact To test if few-shot approach can effectively reduce the amount of training samples to train a plant counting object detector we decided to use the state-of-the-art disputed cross-domain method few-shot model that have shown promising results in the literature. CD-ViTO is an advanced few-shot object detection model that specifically addresses the challenges of domain shift between base and novel classes. The model employs a dual-branch vision transformer backbone that separately processes visual and textual information, with a specialized feature fusion mechanism to integrate these modalities. CD-ViTO's architecture contains a domain-adaptive feature alignment module that minimizes the distribution discrepancy between source and target domains through adversarial learning techniques. Unlike previous few-shot approaches that rely primarily on meta-learning, CD-ViTO leverages cross-domain adaptation strategies to better generalize from base to novel classes with minimal examples (typically 1-10 shots). The backbone consists of a modified ViT (Vision Transformer) structure with domain-invariant attention layers that help extract features robust to domain variations. In benchmark evaluations, CD-ViTO has demonstrated superior performance compared to previous few-shot object detection methods, achieving up to 15% higher AP scores on cross-domain detection tasks and showing particular strength in agricultural applications where domain shifts are common due to variations in growth conditions, illumination, and plant morphology.

2.4. *zero-shots object detectors*

2.5. *Handcrafted object detector*

Many HC object detectors for maize seedlings count have been proposed in the literature. Unlike other works [4] we wrote an HC algorithm to get accurate annotated dataset, sacrificing the amount of images succesfully detected for the quality of the annotations.

This HC algorithm is havelly based on agronomical knowledge and color thresholding. For the execution of this algorithm, the user must set the following parameters:

- HSV minimum and maximum thresholds
- minimum and maxiumum leaf area for plant
- minimum distance between plants on rows
- distance inter rows

In summary, the HC algorithm is based on color thresholding, that rely on the manual setting of Hue, Saturation, and Value (HSV) color minimum and maximum thresholding. Thresholding creates a binary image that is subsequently regionalized by connected components. The connected components are then filtered by area. Centers of the areas are fitted with RANSAC to get the slope of the lines where plants lie. Only the tiles where the slope is within the interquartile range are considered for further processing. The centers of the

filtered tiles are then fitted again with RANSAC to get the lines on the neighboring of the first line detected.

This as most of HC algorithm is intrinsically limited by the color thresholding. Our HC is based on the color of the maize seedlings, which is green.

3. Results

4. Discussion

5. Conclusions

Author Contributions: For research articles with several authors, a short paragraph specifying their individual contributions must be provided. The following statements should be used “Conceptualization, X.X. and Y.Y.; methodology, X.X.; software, X.X.; validation, X.X., Y.Y. and Z.Z.; formal analysis, X.X.; investigation, X.X.; resources, X.X.; data curation, X.X.; writing—original draft preparation, X.X.; writing—review and editing, X.X.; visualization, X.X.; supervision, X.X.; project administration, X.X.; funding acquisition, Y.Y. All authors have read and agreed to the published version of the manuscript.”, please turn to the [CRediT taxonomy](#) for the term explanation. Authorship must be limited to those who have contributed substantially to the work reported.

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Abbreviations

The following abbreviations are used in this manuscript:

- MDPI Multidisciplinary Digital Publishing Institute
- DOAJ Directory of open access journals
- TLA Three letter acronym
- LD Linear dichroism

Appendix A

Appendix A.1

The appendix is an optional section that can contain details and data supplemental to the main text—for example, explanations of experimental details that would disrupt the flow of the main text but nonetheless remain crucial to understanding and reproducing the research shown; figures of replicates for experiments of which representative data are shown in the main text can be added here if brief, or as Supplementary Data. Mathematical proofs of results not central to the paper can be added as an appendix.

Table A1. This is a table caption.

Title 1	Title 2	Title 3
Entry 1	Data	Data
Entry 2	Data	Data

Appendix B

All appendix sections must be cited in the main text. In the appendices, Figures, Tables, etc. should be labeled, starting with “A”—e.g., Figure A1, Figure A2, etc.

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